

## Unit 5, Lesson 7: Solving Problems Involving Rectangles on the Coordinate Plane

### Benchmarks

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same  $x$ -coordinate or the same  $y$ -coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

### Learning Targets

- ☐ I can find distances between ordered pairs.
- ☐ I can find the area of a rectangle on a coordinate plane.
- ☐ I can find the fourth vertex of a rectangle on a coordinate plane.

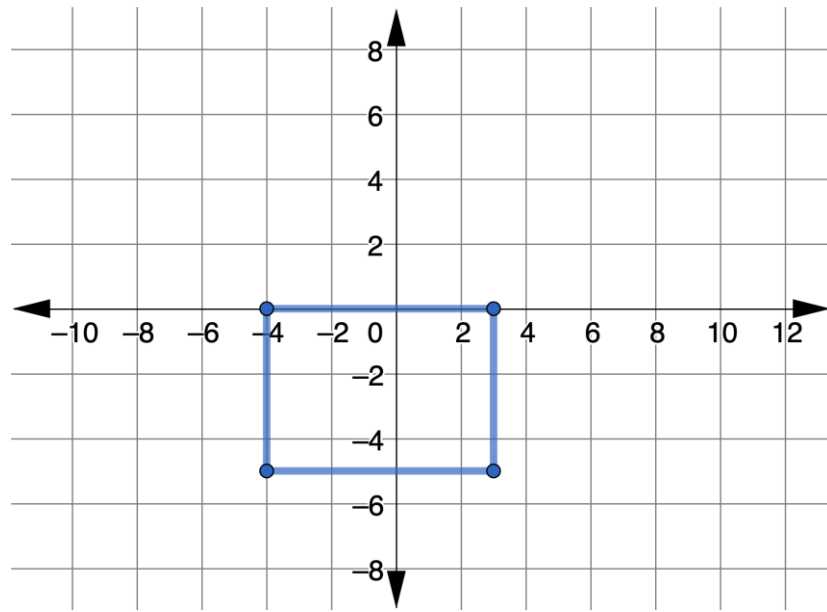
**5.7.1 Warm-Up: True Statement**

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{\phantom{0}}.\boxed{\phantom{0}} \times 3.2 = \boxed{\phantom{0}}.\boxed{\phantom{0}}$$

### 5.7.2 Collaboration: Area of Rectangles

1. Work with your partner to explain how to find the area of a rectangle.
2. A rectangle with coordinates  $(3, 0)$ ,  $(3, -5)$ ,  $(-4, 0)$ , and  $(-4, -5)$  is shown. The scale of the grid is in centimeters (cm).

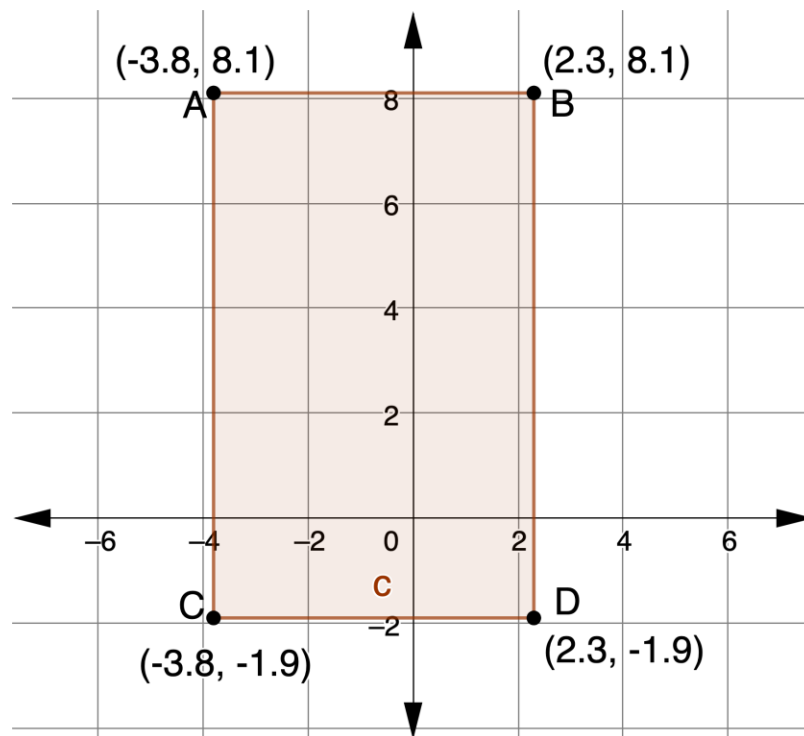


Work with your partner to answer the following questions.

- a. Determine the length and the width of the rectangle. Include units.
- b. What is the area of the rectangle? Include units.

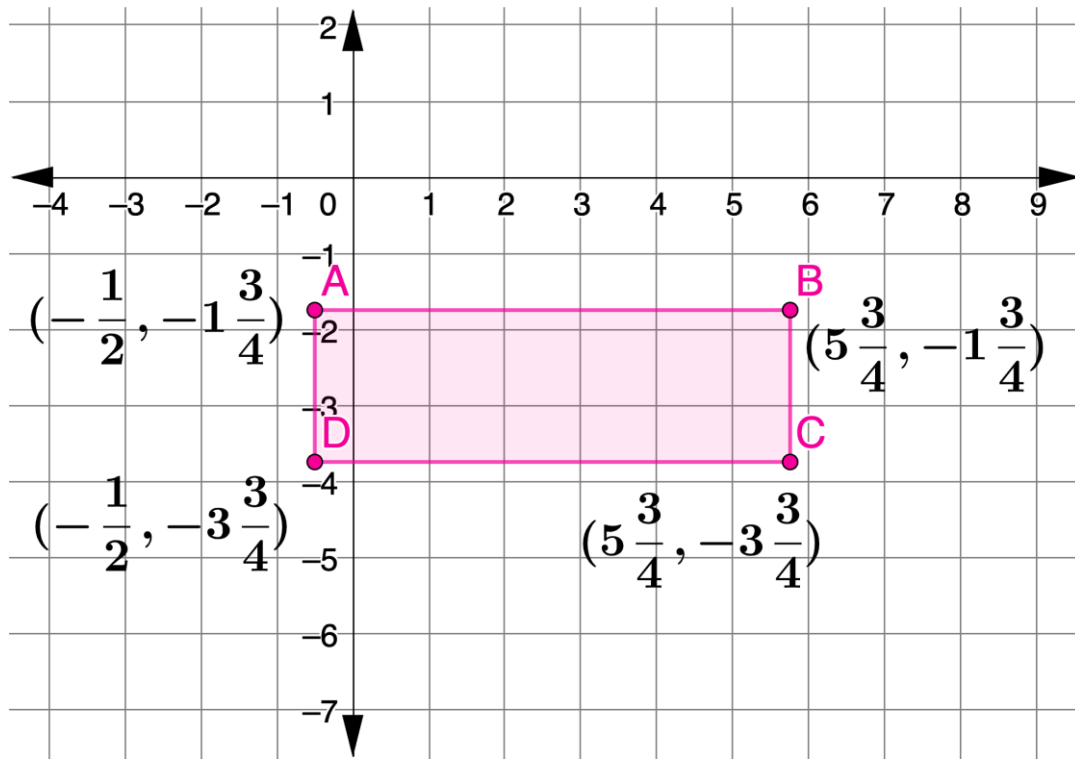
**5.7.3 Guided Practice: Finding the Area of Rectangles on the Coordinate Plane**

1. Henry wants to install new carpet in his living room. A drawing of his living room, with measurements in feet, is shown.



What is the area of the living room?

2. A rectangle is shown.



What is the area of the rectangle?

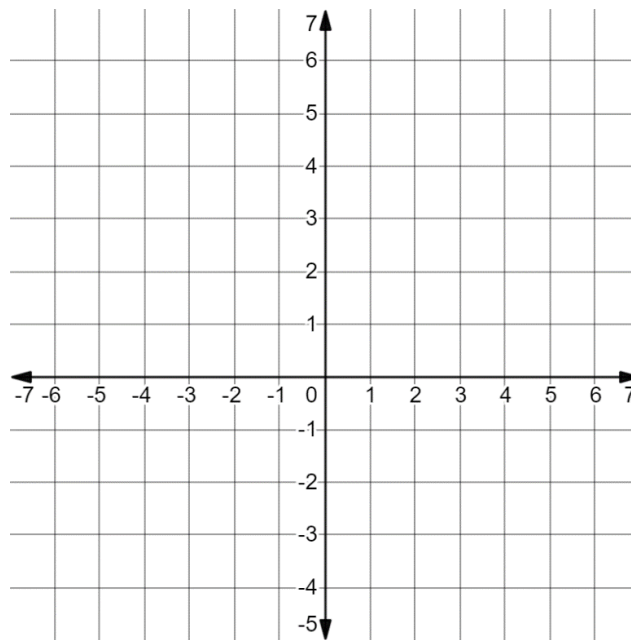
### 5.7.4 Collaboration: Finding the Fourth Vertex

1. With your partner, list every attribute you know about rectangles.

| Rectangles |
|------------|
|            |

2. With your partner, find another pair of partners. Share what you wrote about rectangles. Add any new attributes the other group found that you didn't.

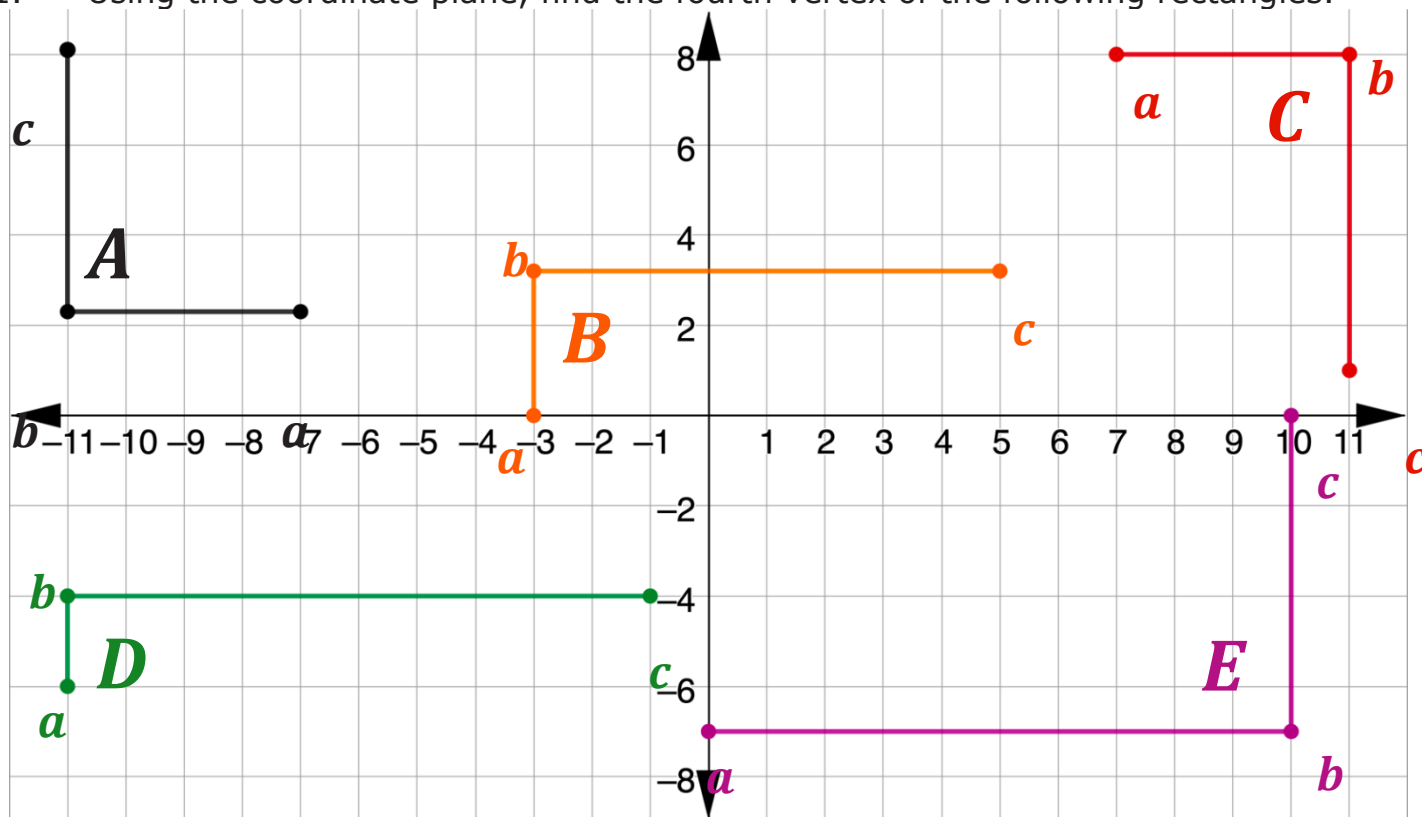
3. On the coordinate grid, draw a rectangle whose sides lie on the gridlines. Label the ordered pair of each vertex.



4. Discuss with your partner how to determine where to place the vertices of your rectangle. Write a summary of your discussion.
5. Based on the rectangle drawn in the previous questions, add any attributes about rectangles to your list.

### 5.7.5 Guided Practice: Finding the Fourth Vertex

1. Using the coordinate plane, find the fourth vertex of the following rectangles.

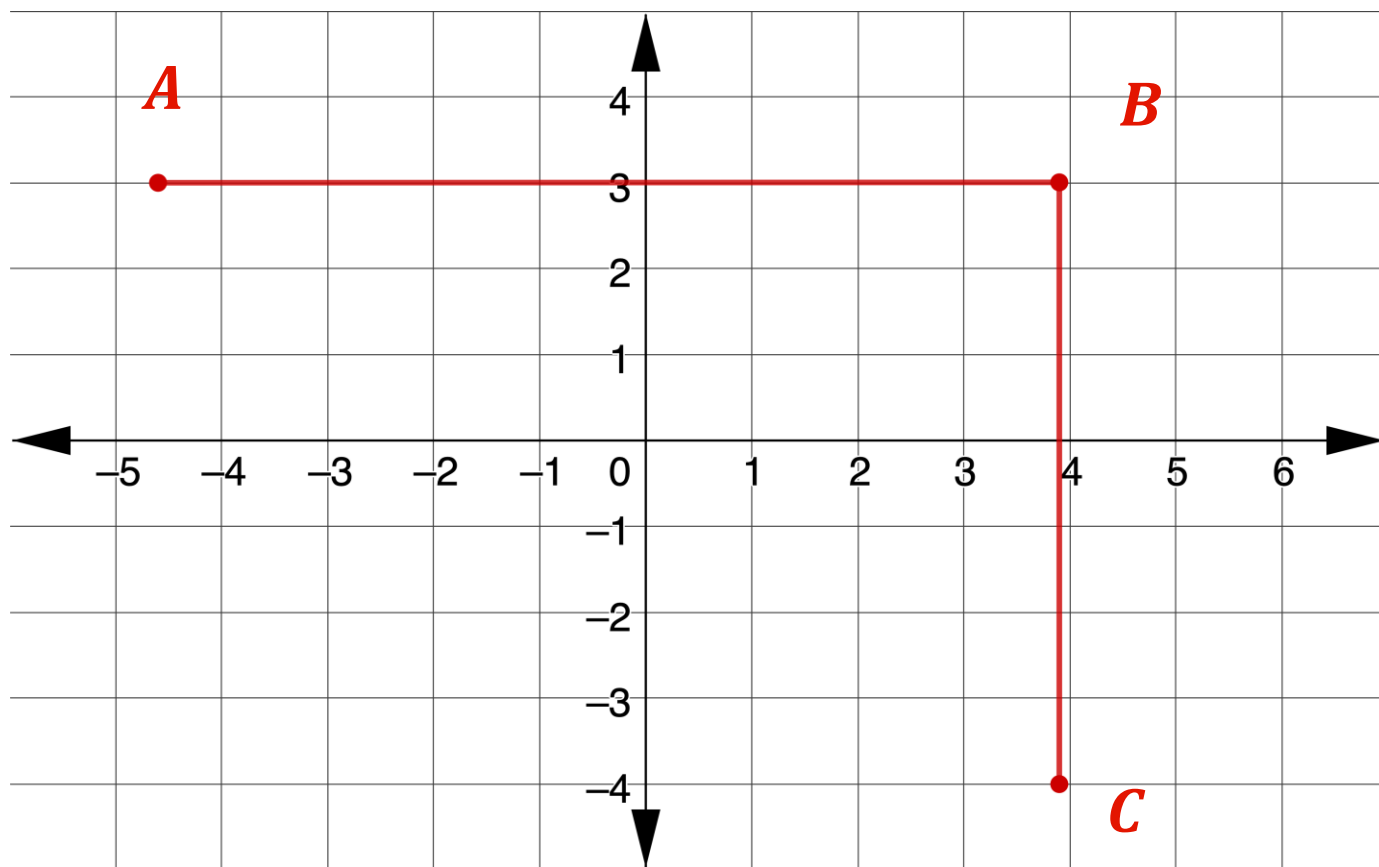


|                    | Vertex <i>a</i> | Vertex <i>b</i> | Vertex <i>c</i> | Vertex <i>d</i> |
|--------------------|-----------------|-----------------|-----------------|-----------------|
| <b>Rectangle A</b> | $(-7, 2.3)$     | $(-11, 2.3)$    | $(-11, 8.1)$    |                 |
| <b>Rectangle B</b> | $(-3, 0.2)$     | $(-3, 3.2)$     | $(5, 3.2)$      |                 |
| <b>Rectangle C</b> | $(7, 8)$        | $(11, 8)$       | $(11, 1)$       |                 |
| <b>Rectangle D</b> | $(-11, -6)$     | $(-11, -4.1)$   | $(-1, -4.1)$    |                 |
| <b>Rectangle E</b> | $(0, -7)$       | $(10, -7)$      | $(10, 0)$       |                 |

2. What is the area of Rectangle A, in square units?

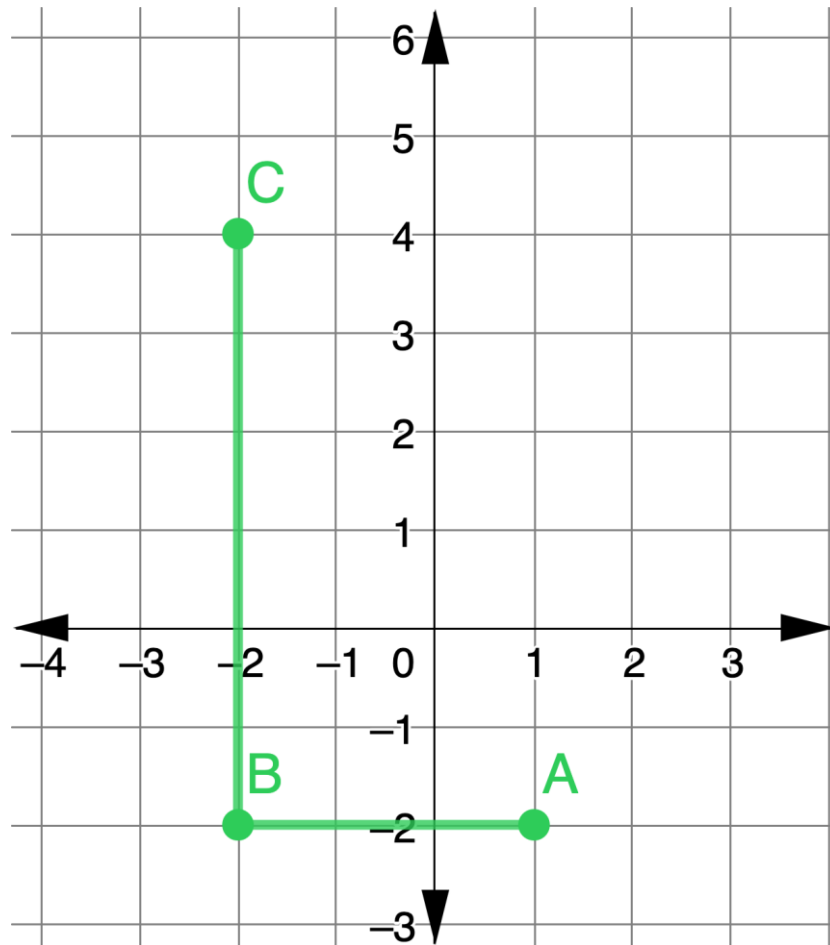
**5.7.6 Your Turn!**

1. Three vertices of a rectangle are at points  $A(-4\frac{3}{5}, 3)$ ,  $B(3\frac{9}{10}, 3)$ , and  $C(3\frac{9}{10}, -4)$ . Each gridline represents 1 centimeter.



- a. What are the coordinates of the fourth vertex?
- b. What is the area of the rectangle?

2. Jada needs to cut out a piece of fabric in the shape of a rectangle. She laid out the fabric on a cutting board, whose scale is represented in feet (ft). Jada placed two edges of the fabric at points  $A$  and  $B$ . She wants the length of the fabric to be as long as the distance from point  $B$  to point  $C$ .



- a. Determine the coordinates of the fourth corner of the fabric.
- b. Find the area of the piece of fabric.

### 5.7.7 Wrap-Up: Reflection

1. Complete the following statements.
  - a. Today, I am still unsure about ...
  - b. To fill this gap, I intend to ...